SYSTEM FOR MAPPING OF FLOOD AND EARLY WARNING CONFIGURATION BY MEANS OF AN UAV

PROJECT NAME:	FLOOD PREDICTION SYSTEM
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WHY THIS PROJECT ?

- Floods cause major loss of life and property and traditional monitoring methods are often slow and limited in coverage. This project aims to provide rapid and accurate flood assessment using UAV technology.
- Early warning through aerial data helps authorities take timely action and improve disaster preparedness.

INTRODUCTION

- This project involves the use of an Unmanned Aerial Vehicle (UAV) and Satellite images to map flood-affected areas and monitor water levels in real time.
- The system collects aerial images and aerial videos to support flood analysis and early warning generation

KEY FEATURES

- Creates a 3D simulation model to give the visual view of the water level.
- Early warning system based on water level data and flood extent analysis.
- Divide the zones for construction areas and provides to Government authorities for Early Planning.

ARCHITECTURE

- The UAV is equipped with cameras, sensors, GPS, and a communication module for data acquisition.
- Collected data is transmitted to a ground station where it is processed for mapping and alert generation.

RESULTS

- The UAV successfully captured high-resolution images of River side areas for accurate mapping of flood.
- The system demonstrated effective early warning capability by detecting rising water levels

FUTURE SCOPE

1. **Integration of AI and machine learning for automatic flood prediction and damage assessment**
2. **Expansion to large-scale deployment with multiple UAVs for continuous monitoring of river basins and urban areas.**